
Output Dilution in MoE Models: Redundant Encoding, Fragile Representations

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Abstract

Mixture-of-Experts (MoE) models appear to encode moral content as robustly as dense models, yet prove far more fragile. In OLMoE-1B-7B, linear probes recover moral valence from nearly every expert-layer combination, with mean peak-layer accuracy above 90%. But these representations collapse under levels of activation noise that a dense model of matched size easily tolerates: a five-fold difference in robustness.

We trace this to *output dilution*. Because the MoE block averages across active experts before contributing to the residual stream, the feedforward signal reaching downstream layers is nearly two orders of magnitude smaller than in a dense MLP. Moral information survives aggregation intact but at a scale trivially overwhelmed by perturbation. Routing itself remains stable under noise; the vulnerability originates entirely in the diluted aggregate.

Checkpoint trajectories confirm this is architectural, not learned: experts never specialize and accuracy saturates within the first few thousand steps. In sparse architectures, redundant encoding does not imply robust encoding.

1 Introduction

Mixture-of-Experts (MoE) architectures route each token through a sparse subset of expert modules, partitioning the representation space into discrete, inspectable units. This structural partition has a natural consequence for alignment research: if moral features concentrate in specific experts, MoE models offer intervention points (expert pruning, expert-specific fine-tuning, router modification) that dense models lack. Conversely, if moral features distribute uniformly across experts, MoE and dense architectures are equivalent for alignment purposes, and the additional complexity of expert-level analysis buys nothing.

We test this question on OLMoE-1B-7B (Muennighoff et al., 2024), a 64-expert, top-8 MoE language model with 6.9B total parameters (1.3B active per token), using the moral probing and fragility methodology from companion work on dense OLMo models (Reblitz-Richardson, 2026). OLMoE is uniquely positioned for this analysis: it is the open MoE model with by far the densest published checkpoint record (244 checkpoints at 5,000-step intervals), and its dense counterpart OLMo-2 1B (from the same lab, with comparable active parameter count and full checkpoint access) gives a controlled architectural comparison.

We report four findings that converge on a single mechanism:

Finding 1: MoEs do not create expert moral specialization. Nearly all 1,024 per-expert probes (64 experts $\times$ 16 layers) decode moral content well above chance (1,020 exceed 75%). At the peak layer, every expert individually exceeds 84% accuracy. The Gini coefficient of expert accuracy is below 0.03 at all layers; moral encoding is as uniformly distributed across experts as it is across neurons in a dense model. The router shows negligible moral content preference (maximum 1.8%).

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Finding 2: MoE encoding is $4.2\times$ more fragile than dense. Despite matching dense OLMo-2 1B on probing accuracy (99.0% vs. 99.0% peak), OLMoE’s moral encoding collapses under $4.2\times$ less noise (mean critical $\sigma^* = 0.92$ vs. 3.81). The fragility gap is not explained by weaker individual expert representations or unstable routing; both are robust in isolation.

Finding 3: The fragility originates in output dilution. The MoE block’s aggregated output (a top-8 weighted average of 64 expert outputs) contributes to the residual stream at $74\times$ smaller scale than the dense MLP output, measured as the standard deviation of the feedforward block’s output across inputs. This *output dilution* means that the same absolute noise level overwhelms the MoE moral signal while leaving the dense signal intact.

Finding 4: Specialization never emerges during training. Across 11 checkpoints spanning OLMoE’s training (step 5K to step 1.2M, covering 20B to 5,033B tokens), the Gini coefficient at the peak layer stays between 0.012 and 0.018 at every checkpoint (cf. 0.016–0.023 across all layers in the final model, §4.2). Moral encoding is present from the earliest available checkpoint (92.1% peak accuracy at step 5K) and remains stable at 93.7% by step 1.2M without ever concentrating in specific experts. The top-5 experts by accuracy change between adjacent checkpoints at near-random rates (Jaccard ≈ 0.09).

The output dilution finding has direct implications for the interpretability of probing accuracy as an alignment metric. Two models can produce identical probing accuracy profiles (high accuracy from early layers, broad encoding across the full network) while differing by nearly two orders of magnitude in the robustness of the underlying signal. Probing accuracy measures what information is *present*; fragility testing, as developed in companion work (Reblitz-Richardson, 2026), measures how *securely* that information is encoded. In MoE architectures, the gap between these two metrics is dramatically larger than in dense models, because the sparse aggregation bottleneck preserves information content while reducing signal scale.

The paper contributes the first expert-level moral probing analysis of an MoE language model, the first quantification of the MoE output dilution effect and its relationship to representational fragility, and a controlled dense-vs-MoE comparison on identical probing methodology. All experiments run on a single MacBook Pro M4 Pro (24 GB, MPS) on base (non-instruct) models.

2 Related Work

Mixture-of-Experts architectures. Sparse MoE was introduced by Shazeer et al. (2017) and scaled by Fedus et al. (2022) and Lepikhin et al. (2021). Recent open MoE models include Mixtral (Jiang et al., 2024), DeepSeek-MoE (Dai et al., 2024), and OLMoE (Muennighoff et al., 2024). OLMoE is unique in publishing 244 training checkpoints, enabling trajectory analysis unavailable for other MoE models.

Expert specialization. Prior work on what individual MoE experts learn has focused on linguistic features (syntax, part-of-speech), domain features (code vs. natural language), and language-specific specialization in multilingual models. Zuo et al. (2022) find that Switch Transformer experts partially specialize by token type. Chi et al. (2022) study expert utilization patterns. To our knowledge, no prior work examines whether MoE experts specialize for moral or ethical features.

Moral probing in language models. Probing classifiers (Belinkov, 2022, Conneau et al., 2018) train lightweight classifiers on model-internal representations to test what information is encoded. Moral probing specifically applies this methodology to moral reasoning features, grounded in Moral Foundations Theory (Graham et al., 2013, Haidt, 2012). Companion work (Reblitz-Richardson, 2026) develops the layer-wise moral probing and fragility testing methodology we extend to MoE models, establishing that fragility resolves structure after probing accuracy saturates.

Activation perturbation and representational robustness. Gaussian noise injection for probing robustness relates to work on representation stability (Morcos et al., 2018) and activation perturbation for identifying causally relevant features (Meng et al., 2022, Vig et al., 2020). Our fragility protocol (Reblitz-Richardson, 2026) adapts this approach to alignment-relevant features, defining critical noise as a quantitative robustness metric.

Dense-model moral encoding. The companion paper (Reblitz-Richardson, 2026) establishes that dense OLMo models encode moral features from early layers (low encoding depth), broadly across

the network (high encoding breadth), with a fragility gradient that continues to resolve after probing accuracy saturates. Prior work on this project also showed that probe-direction suppression in dense 1B models does not capture behavior due to feature redundancy, which motivated investigating whether MoE’s structural partition reduces this redundancy.

OLMo ecosystem. OLMo (Groeneveld et al., 2024) and OLMoE (Muennighoff et al., 2024) are developed by the Allen Institute for AI with a commitment to open science, including full training data, code, intermediate checkpoints, and evaluation infrastructure. This openness enables the controlled architectural comparison (§4.1), the output scale measurement (§4.4), and the 11-checkpoint trajectory analysis (§4.5) that are central to our findings.

3 Methodology

3.1 Models and Comparison Design

OLMoE-1B-7B (allenai/OLMoE-1B-7B-0924; Muennighoff et al., 2024) is a 16-layer MoE language model with 64 experts per layer, top-8 routing, 6.9B total parameters (1.3B active per token), and hidden dimension 2048. Each expert is a gated MLP with intermediate dimension 1024, using SiLU activation. The router is a learned linear projection ($2048 \rightarrow 64$) followed by softmax and top- k selection with normalized weights. The model is trained with a load-balancing auxiliary loss ($\lambda = 0.01$) to encourage uniform expert utilization.

OLMo-2 1B (allenai/OLMo-2-0425-1B; OLMo Team, 2025) is a 16-layer dense transformer with 1.5B parameters and hidden dimension 2048. It serves as the architectural control: same lab, same training philosophy, comparable active parameter count, same number of layers and hidden dimension.

Both models are base (non-instruct) checkpoints. All experiments use the same 240-pair moral probing dataset (§3.5), the same probe architecture (§3.3), and the same fragility protocol (§3.4). Architecture is the independent variable.

3.2 Per-Expert Activation Collection

Standard layer-wise probing (Reblitz-Richardson, 2026) registers forward hooks on transformer layer outputs to collect post-layer hidden states. For per-expert probing, we bypass the router and compute all 64 expert outputs in parallel.

For each input text, we hook `post_attention_layernorm` at each layer to capture the pre-MoE hidden state $h \in \mathbb{R}^{s \times d}$ (where s is sequence length, $d = 2048$). We then compute expert outputs by directly applying each expert’s FFN weights to the mean-pooled hidden state $\bar{h} = \frac{1}{s} \sum_t h_t$:

$$\text{gate_up}_e = \bar{h} \cdot W_e^{\text{gate_up}\top} \in \mathbb{R}^{2k}$$

$$g_e, u_e = \text{chunk}(\text{gate_up}_e) \in \mathbb{R}^k$$

$$o_e = \text{SiLU}(g_e) \odot u_e \cdot W_e^{\text{down}\top} \in \mathbb{R}^d$$

where $k = 1024$ is the intermediate dimension and $e \in \{0, \dots, 63\}$. This computation is batched across all 64 experts using `torch.einsum`, yielding all expert outputs in a single operation per layer.

For router analysis, we also capture the router logits by computing $\bar{h} \cdot W^{\text{gate}\top} \in \mathbb{R}^{64}$, where W^{gate} is the router’s learned weight matrix.

Clean aggregated output. To produce the MoE block’s actual output for downstream probing, we apply the standard routing: softmax over router logits, select top-8, normalize weights, and compute the weighted sum of the selected experts’ outputs.

3.3 Probing Architecture

All probes are binary linear classifiers: `nn.Linear(d, 1)` trained with binary cross-entropy loss, Adam optimizer ($\text{lr} = 10^{-2}$), 50 epochs. For layer-level probes, $d = 2048$ (full hidden dimension). For per-expert probes, $d = 2048$ (expert output dimension, which equals hidden dimension in OLMoE’s architecture). For aggregated-MoE probes used in the perturbation experiments, $d = 2048$.

The probe threshold is 0 (logit sign determines classification). Accuracy is reported on a held-out test set (48 pairs, 96 texts).

3.4 Fragility Protocol

We extend the fragility testing protocol from companion work (Reblitz-Richardson, 2026). In the standard protocol, Gaussian noise $\mathcal{N}(0, \sigma^2 I)$ is injected into post-layer hidden states at magnitudes $\sigma \in \{0.1, 0.3, 1.0, 3.0, 10.0\}$, and the critical noise σ^* is the smallest σ at which probe accuracy drops below 0.6.

For the MoE component perturbation experiment (§4.4), we extend this protocol to three perturbation targets within the MoE block:

1. **Router perturbation.** Noise is added to the router logits before softmax and top- k selection. This changes both which experts are selected and their aggregation weights.
2. **Expert perturbation.** Noise is added to individual expert outputs before weighted aggregation. Routing is held fixed (clean logits determine expert selection and weights).
3. **Output perturbation.** Noise is added to the final aggregated MoE output (control condition equivalent to standard fragility testing).

For each condition, probes are trained on clean aggregated outputs (train set) and evaluated on perturbed outputs (test set) at each noise level. Results are averaged over 10 random seeds per noise level to reduce variance from individual noise realizations.

Output scale measurement. To interpret the component fragility results, we measure the natural scale of each component (standard deviation across test texts) and the feedforward output scale at each layer for both OLMoE and OLMo-2 on the same input texts.

3.5 Probing Dataset

We use the same 240-pair moral probing dataset as companion work (Reblitz-Richardson, 2026): 40 minimal pairs per Moral Foundations Theory foundation (care/harm, fairness/cheating, loyalty/betrayal, authority/subversion, sanctity/degradation, liberty/oppression), subsampled with a deterministic seed from a 1,200-pair dataset constructed per published quality guidelines with LLM-assisted filtering for naturalness and moral neutrality of neutral-side sentences (see DATASET_GUIDELINES.md). The subsample is split 80/20 into 192 training pairs (384 texts) and 48 test pairs (96 texts), with foundation balance preserved across splits.

Dataset identity is load-bearing: the dense-vs-MoE comparison (§4.1) and the output scale comparison (§4.4) use identical inputs to ensure any observed differences are architectural, not data-driven.

3.6 Checkpoint Trajectory Analysis

OLMoE publishes 244 training checkpoints at 5,000-step intervals from step 5,000 (20B tokens) through step 1,220,000 (5,117B tokens). We select 11 checkpoints spanning training: dense early sampling (steps 5K, 10K, 20K, 50K, 100K) and logarithmic spacing through the remainder (steps 200K, 400K, 600K, 800K, 1M, 1.2M). Our sample therefore ends at step 1,200,000 (5,033B tokens), just short of the published set’s final step 1,220,000 (5,117B tokens); the 84B-token gap does not affect any trajectory conclusion. At each checkpoint, we run the full per-expert probing analysis (§3.2–3.3) and router analysis (§3.2), computing the Gini coefficient of per-expert moral accuracy and tracking expert identity stability (Jaccard similarity of the top-5 experts between adjacent checkpoints).

Each checkpoint is loaded sequentially (load, probe, free) to fit within 24 GB memory. Results are saved per-checkpoint with resume support, enabling interrupted runs to continue from the last completed checkpoint.

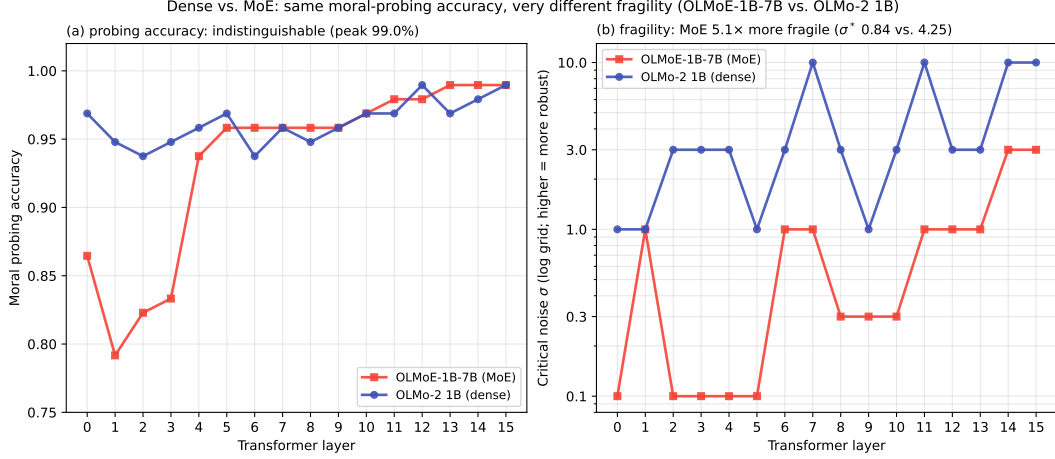


Figure 1: Dense and MoE encode moral content with near-identical accuracy but very different robustness. (a) Per-layer moral probing accuracy for OLMoE-1B-7B and dense OLMo-2 1B; both peak at 99.0%, differing only at the early layers. (b) Per-layer critical noise σ^* (smallest σ at which probe accuracy falls below 0.6, on the log grid $\{0.1, 0.3, 1.0, 3.0, 10.0\}$): OLMoE is $5.1\times$ more fragile (mean σ^* 0.84 vs. 4.25) and concentrates robustness in the final two layers.

3.7 Hardware and Reproducibility

All experiments run on a MacBook Pro M4 Pro (24 GB unified memory) using PyTorch MPS backend with float16 precision. OLMoE-1B-7B requires ~ 14 GB in float16; OLMo-2 1B requires ~ 3 GB. Models are loaded sequentially (load, evaluate, free) to fit within memory.

A monkey-patch to `torch.histc` is required for OLMoE on MPS: the MoE router’s token-counting operation uses integer `histc`, which is not implemented on MPS or CPU. The patch casts to float and falls back to CPU for this single operation.

All random seeds, model revisions, and command-line invocations are recorded in the output JSON files. Experimental scripts are available at `papers/2_moe_output_dilution/scripts/`.

4 Results

4.1 Dense vs. MoE: Same Accuracy, Different Robustness

We first establish the baseline comparison between OLMoE-1B-7B and dense OLMo-2 1B using the standard layer-wise moral probing and fragility battery from companion work (Reblitz-Richardson, 2026). Both models have 16 transformer layers and comparable active parameter counts (1.3B active for OLMoE vs. 1.5B for OLMo-2), enabling a controlled architectural comparison on the same 240-pair probing dataset.

Probing accuracy is indistinguishable. OLMoE achieves peak probing accuracy of 99.0% at layer 13; OLMo-2 achieves 99.0% at layer 12. Both models reach onset (accuracy > 0.6) at layer 0 and maintain encoding breadth of 1.0, meaning moral content is decodable from every layer. The probing accuracy profiles differ only in that OLMoE shows lower early-layer accuracy (79–86% at layers 0–3 vs. 94–97% for OLMo-2) before converging at later layers.

Fragility diverges sharply. Under Gaussian noise injection at $\sigma \in \{0.1, 0.3, 1.0, 3.0, 10.0\}$ averaged over 10 noise seeds, OLMoE is $4.2\times$ more fragile than OLMo-2: mean critical noise $\sigma^* = 0.92$ vs. 3.81. The fragility profiles also differ structurally. OLMo-2 shows distributed robustness, with critical noise ≥ 3.0 at 12 of 16 layers and ≥ 10.0 at layers 13, 14, and 15. OLMoE concentrates robustness in the final three layers only (critical noise 3.0 at layers 13–15; ≤ 0.3 at 9 of 16 layers, and ≤ 1.0 at 13 of 16).

Figure 1 contrasts the two architectures across both metrics.

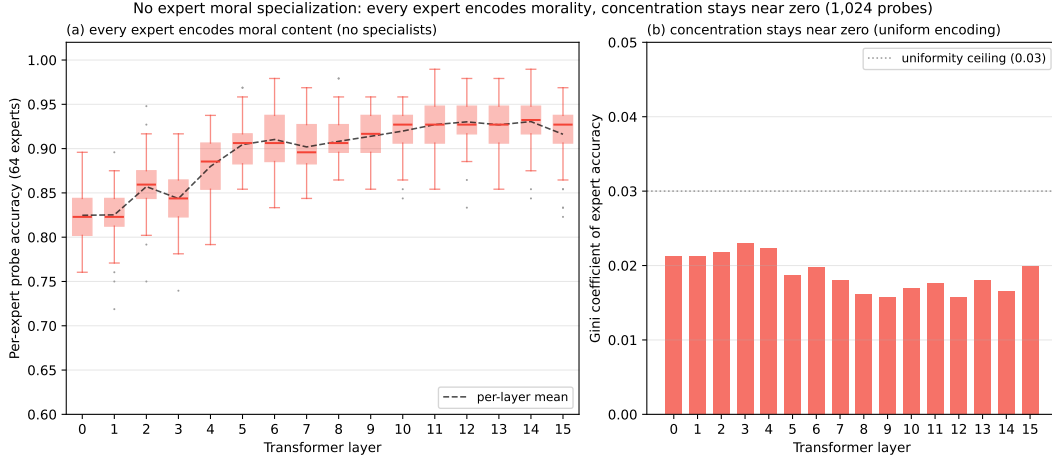


Figure 2: No expert moral specialization. (a) Distribution of the 64 per-expert probe accuracies at each layer (box plot) with the per-layer mean overlaid; every expert encodes moral content, with no sparse high-accuracy subset. (b) The per-layer Gini coefficient of expert accuracy stays in $[0.016, 0.023]$, far below any concentration threshold, and is lowest at the late layers where encoding peaks.

This establishes the puzzle the remaining experiments investigate: both architectures encode moral content with near-identical accuracy, but the MoE encoding is substantially more fragile. What is it about MoE that produces this gap?

4.2 No Expert Moral Specialization

We trained 1,024 independent binary probes, one per expert-layer combination ($64 \text{ experts} \times 16 \text{ layers}$), on per-expert activations collected by bypassing the router and computing all 64 expert FFN outputs in parallel via batched einsum on the pre-MoE hidden state. If MoE architectures create expert-level moral specialization, we would expect a sparse subset of experts to achieve high probe accuracy while most remain near chance.

The result is the opposite: moral encoding is uniformly distributed across all experts at every layer. 1,020 of 1,024 expert probes exceed 75% accuracy (four exceptions at early layers 1–3, ranging from 72–75%). At the peak layer (layer 14), all 64 experts individually exceed 84% accuracy (mean 93.0%, min 84.4%). The per-layer Gini coefficient of expert accuracy, which measures how concentrated moral signal is across experts, ranges from 0.016 to 0.023, indicating near-perfect uniformity. Gini is modestly higher in early layers (0.021–0.023 at layers 0–3) and lowest in mid-network (0.016 at layers 8, 9, and 12), suggesting that moral encoding becomes *more* uniform through the early and middle layers.

Figure 2 shows the per-expert accuracy distribution at every layer.

This finding has immediate consequences for alignment interventions. Dense models encode moral features diffusely across neurons within each layer; MoE partitions representations across 64 discrete expert modules, yet moral features remain equally diffuse across all 64. The structural partition MoE introduces does not induce functional specialization for moral content.

4.3 Router Is Content-Agnostic for Morality

The absence of expert specialization raises the question of whether the router treats moral and neutral inputs differently. We analyzed per-layer routing distributions by comparing mean router probabilities and top-8 selection frequencies conditioned on moral vs. neutral input texts.

The router shows negligible moral preference. The maximum routing preference (the largest absolute difference in mean routing probability between moral and neutral inputs for any single expert) is 1.8% (layer 12, expert 37). Using a threshold of 0.5% absolute routing-probability difference, the number of experts with any detectable preference ranges from 3 (layer 1) to 19 (layer

6) out of 64, but all preference magnitudes are small: the 95th percentile across all 1,024 expert-layer combinations is below 2%.

Combined with §4.2, this establishes that moral encoding in OLMoE is doubly diffuse: the router does not segregate moral tokens to specific experts, and every expert that receives tokens encodes moral content with comparable accuracy. MoE and dense architectures produce equivalent moral encoding geometry despite their structural differences.

4.4 Output Dilution Explains MoE Fragility

Having established that moral encoding is uniformly distributed across experts and that the router is content-agnostic, we turn to the source of the $4.2\times$ fragility gap. We isolated three perturbation targets within the MoE block:

- **Router perturbation:** Gaussian noise on router logits before softmax and top- k selection, changing which experts are selected and their aggregation weights.
- **Expert perturbation:** Gaussian noise on individual expert outputs before weighted aggregation.
- **Output perturbation:** Gaussian noise on the final aggregated MoE output (control condition matching §4.1).

For each condition, probes were trained on clean aggregated MoE outputs and evaluated on perturbed outputs at noise levels $\sigma \in \{0.01, 0.03, 0.1, 0.3, 1.0, 3.0, 10.0\}$, averaged over 10 random seeds.

The component fragility ranking reverses the natural hypothesis. The router is the *most robust* component: 8 of 16 layers never reach the fragility threshold at any tested noise level. Following the cap-at-maximum convention (§3.4) those never-fragile layers are censored at $\sigma = 10$ rather than dropped, giving mean critical noise $\sigma^* = 9.56$ (it is 9.13 if they are dropped instead). Expert outputs are moderately fragile ($\sigma^* = 1.8$, all 16 layers fragile). The aggregated output is the most fragile ($\sigma^* = 0.6$, all 16 layers fragile), consistent with the full-hidden-state fragility from §4.1.

This counterintuitive ranking is explained by the natural scales of each component. The MoE block’s aggregated output has a standard deviation of only 0.003–0.008 at layers 0–8, orders of magnitude smaller than the router logit scale (~ 0.5) and comparable to the smallest tested noise levels.

4.4.1 The $74\times$ output scale gap

To test whether this small output scale is an inherent property of MoE aggregation, we directly measured the feedforward output scale at every layer for both OLMoE and OLMo-2 on the same 100 input texts. The dense MLP produces outputs **$74\times$ larger on average** than the MoE block, measured as the standard deviation of the mean-pooled feedforward output across texts:

Layer	OLMoE MoE std	OLMo MLP std	Ratio
0	0.003	0.448	$167\times$
5	0.003	0.291	$108\times$
8	0.008	0.459	$60\times$
12	0.018	1.070	$61\times$
15	0.096	8.779	$91\times$

The ratio exceeds $60\times$ at 9 of 16 layers. The per-layer ratio varies widely, from $5.3\times$ (layer 2, where the MoE output scale spikes) to $167\times$ (layer 0); the table above excerpts five representative layers. The MoE block’s contribution to the residual stream is not just smaller; it operates on a fundamentally different scale than the dense MLP.

Figure 3 relates the output-scale gap to component fragility.

This output dilution is the mechanism behind MoE fragility. Because only 8 of 64 experts contribute to each token’s MoE output, and the routing weights further attenuate each expert’s contribution, the MoE block injects a much smaller perturbation into the residual stream than a dense

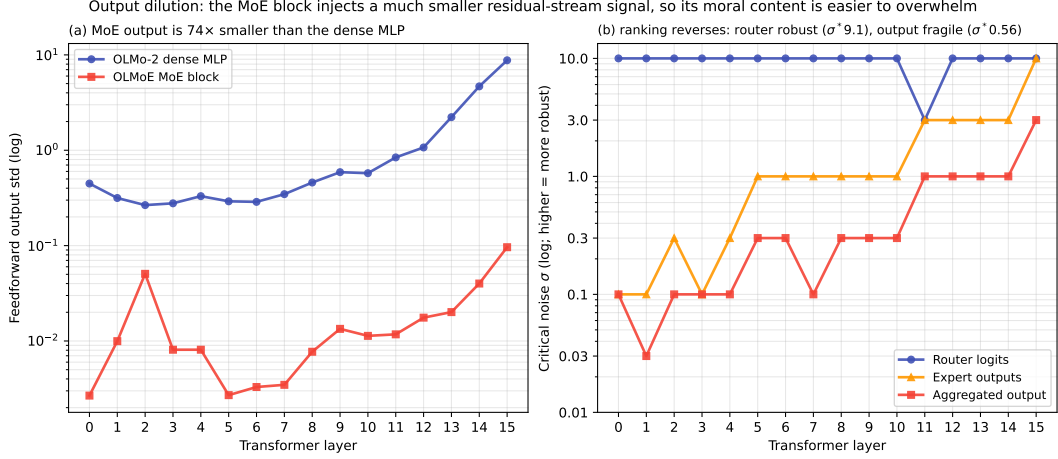


Figure 3: Output dilution explains MoE fragility. (a) Per-layer feedforward output scale (standard deviation of the mean-pooled output) for the OLMo MoE block vs. the dense OLMo-2 MLP; the dense MLP output is $74\times$ larger on average. (b) Per-layer critical noise for the three MoE perturbation targets: the router is most robust (mean $\sigma^* 9.1$), the aggregated output most fragile (mean $\sigma^* 0.56$), because the output operates on the diluted scale from panel (a).

MLP. The moral signal carried by this small perturbation is correspondingly easier to overwhelm with noise.

The finding cleanly connects all four prior results:

1. **Probing accuracy is preserved** (§4.1) because the MoE output, though small, contains the same information content as the dense MLP output; a linear probe with learned weights can amplify the signal.
2. **No expert specialization** (§4.2) because every expert processes the same pre-MoE hidden state and applies the same architectural pattern; specialization would require the router to route moral content selectively, which it does not (§4.3).
3. **Fragility increases** (§4.1) because the absolute noise threshold to disrupt a 0.003-scale signal is much lower than for a 0.3-scale signal.
4. **Router robustness** (§4.4) because the routing mechanism operates on logits at scale ~ 0.5 , far above the noise levels that disrupt the MoE output.

4.5 Specialization Never Emerges During Training

OLMoE publishes 244 training checkpoints at 5,000-step intervals, spanning from step 5,000 (20B tokens) to step 1,220,000 (5,117B tokens). We ran the per-expert probing analysis (§4.2) and router analysis (§4.3) at 11 checkpoints spanning training: dense early sampling (steps 5K, 10K, 20K, 50K, 100K) and logarithmic spacing through the remainder (steps 200K, 400K, 600K, 800K, 1M, 1.2M).

Moral encoding appears from the earliest available checkpoint. At step 5,000 (20B tokens, $\sim 0.4\%$ of training), per-expert mean accuracy already reaches 92.1% at the peak layer, with 1,006 of 1,024 expert probes above 75%. Accuracy is remarkably stable throughout training (93.6% at step 10K, 93.2% at step 200K, 93.7% at step 1.2M), fluctuating in a narrow 92–94% band rather than progressively sharpening. The peak layer stabilizes at layer 14 from step 200K onward, matching the final model’s peak.

Specialization never appears at any checkpoint. The Gini coefficient of per-expert accuracy remains between 0.012 and 0.018 at the peak layer across all 11 checkpoints, never exceeding 0.03 at any layer of any checkpoint. The trajectory plot shows accuracy stable while Gini stays flat: the model maintains uniform moral representations throughout training without concentrating them in specific experts. Overall mean Gini (averaged across all 16 layers) shows a mild *decrease* from 0.020 at step 50K to 0.018 at step 1M, suggesting that training produces more *uniform* encoding, not more specialized.

Specialization never emerges: accuracy stable and concentration flat across training (11 checkpoints, 20B → 5,117B tokens).

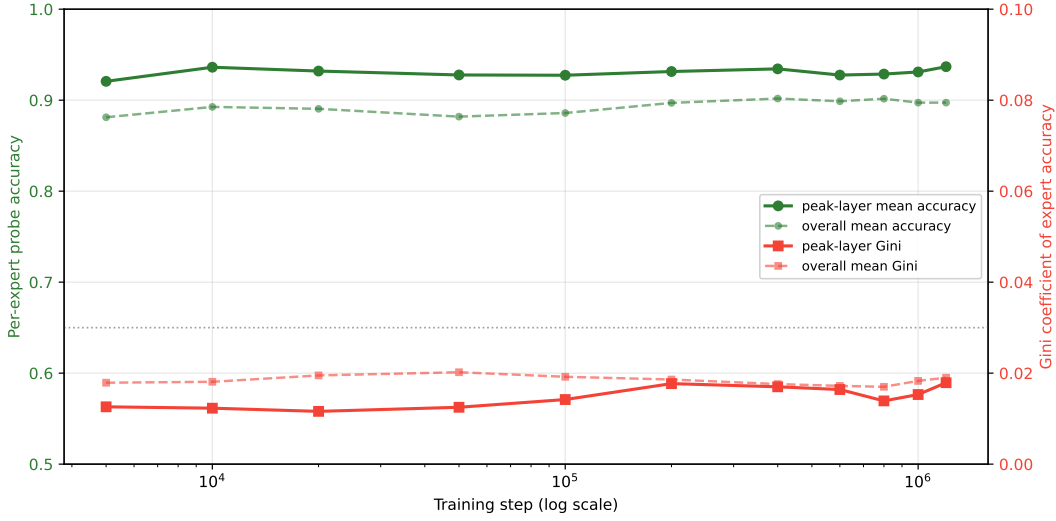


Figure 4: Specialization never emerges during training. Across 11 OLMoE checkpoints (step 5K–1.2M, 20B–5,117B tokens), peak-layer and overall mean per-expert accuracy (left axis) stay in a 92–94% band from the earliest checkpoint, while the Gini coefficient of expert accuracy (right axis) stays flat near zero. Moral encoding is present from the start and never concentrates into specific experts.

Figure 4 plots the training trajectory of accuracy and concentration.

Expert identity is unstable. The Jaccard similarity of the top-5 highest-accuracy experts between adjacent checkpoints fluctuates near the random baseline of $5/64 \approx 0.08$, ranging from 0.0 (complete turnover) to 0.25 (two shared experts out of five). No stable “moral expert” identity exists; the ranking of experts by moral accuracy is noise around a uniform mean, not a consistent specialization pattern.

5 Discussion

5.1 Output Dilution as an Architectural Property

The $74\times$ output scale gap between MoE and dense feedforward blocks is not specific to moral encoding; it is a structural consequence of sparse expert aggregation. When a top- k routing mechanism selects 8 of 64 experts, each expert contributes roughly $\frac{1}{8}$ of the aggregated output (modulated by routing weights). The aggregated output is therefore a weighted average of 8 expert outputs, each operating on a 1024-dimensional intermediate space, producing a 2048-dimensional output. The dense MLP, by contrast, applies its full parameter budget to every token, producing a larger-scale output.

The dilution effect likely scales with the sparsity ratio. OLMoE uses top-8 of 64 (12.5% sparsity); models with higher sparsity (e.g., top-2 of 8 in Mixtral) may show even stronger dilution, while lower sparsity would reduce it. The load-balancing auxiliary loss, which encourages uniform expert utilization, may further contribute by preventing any single expert from dominating the aggregated output.

An important caveat: the output dilution we measure is the *feedforward block’s contribution to the residual stream*, not the total signal in the residual stream. Because the residual connection carries forward the pre-MoE hidden state, the full hidden state after the MoE block is dominated by the residual, not the MoE output. This means that probing the full hidden state (as in §4.1) picks up moral signal from both the MoE contribution and accumulated residual contributions from earlier layers. The fragility difference arises because noise added to the full hidden state disrupts the MoE contribution disproportionately: the noise is small relative to the residual but large relative to the MoE output.

5.2 Implications for Alignment Interventions

The absence of expert moral specialization (§4.2) closes one potential intervention path: there are no “moral experts” to prune, fine-tune, or monitor. MoE architectures, despite their structural partition into discrete expert modules, do not make moral encoding more tractable for targeted intervention than dense models do.

The output dilution finding (§4.4), however, opens a different concern. If moral features in MoE models are encoded at very small scale in the feedforward contribution, they may be easier to accidentally destroy during fine-tuning. A LoRA adapter that modifies the feedforward weights by even a small amount in absolute terms could represent a large *relative* perturbation to the MoE output. This prediction is testable: companion work’s C15 finding (fragility-locus shift under insecure-code LoRA in dense OLMo) could be replicated on OLMoE, where we would predict a larger fragility shift from the same fine-tuning recipe.

5.3 Feature Redundancy Across Architectures

Prior work on this project (Reblitz-Richardson, 2026) found that probe-direction suppression in dense 1B models does not capture behavior: a gradient penalty suppresses the probe direction by 3.07 SD with no effect on behavioral judge scores (within 0.01 / 10). This was attributed to feature redundancy: at the 1B scale, the model has enough representational capacity to encode persona features along directions orthogonal to the probe’s extracted direction.

The present findings show that MoE architecture does not resolve this redundancy problem. Despite partitioning representations across 64 discrete modules, moral features are encoded equally strongly in every module. The structural partition of MoE is orthogonal to the functional organization of moral features; the model encodes the same information in every expert, just as a dense model encodes it across every neuron.

The checkpoint trajectory analysis (§4.5) strengthens this conclusion: the absence of specialization is not a late-training convergence but is present from step 5K (20B tokens), before the routing mechanism has fully matured. Training does not create and then destroy expert moral specialization; it never exists.

Feature redundancy in language models is therefore not a consequence of architectural homogeneity (all neurons participating in everything) but of training dynamics: the training objective distributes useful features across all available representational capacity, regardless of how that capacity is architecturally partitioned. Load-balancing losses in MoE, which encourage uniform expert utilization, may actively reinforce this tendency.

5.4 Probing Accuracy as an Alignment Metric

The dense-vs-MoE comparison starkly illustrates the insufficiency of probing accuracy as a standalone alignment metric. Both models achieve near-perfect probing accuracy (99–100%) with full encoding breadth (decodable at every layer) and near-zero encoding depth (onset at layer 0). On probing accuracy alone, the two architectures are indistinguishable. Yet the MoE model’s moral encoding is $4.2\times$ more fragile, and the underlying feedforward signal is $74\times$ weaker.

Probing accuracy measures the *presence* of information: whether a linear classifier can extract a feature from the representation. Fragility testing measures the *security* of that information: how much perturbation the encoding can withstand before the feature becomes unextractable. The output dilution mechanism shows that these two metrics can diverge dramatically: information can be present (high accuracy) but insecure (low fragility), encoded at a scale that is trivially disrupted.

The early-layer accuracy gap between architectures offers a second diagnostic. OLMoE’s early layers (0–3) achieve only 79–86% per-expert accuracy, compared to 94–97% for OLMo-2 at the same layers. When the probing dataset was tightened to remove superficial cues (animacy and register confounds; see §3.1 of Reblitz-Richardson (2026)), OLMoE’s early-layer accuracy dropped more than OLMo-2’s, suggesting that the MoE architecture’s diluted output makes early layers more dependent on shallow features. This is consistent with output dilution: when the feedforward contribution to the residual stream is small, early layers cannot inject enough signal to support robust classification, and probes compensate by exploiting dataset artifacts when available.

This reinforces the methodological argument from companion work that fragility testing is a necessary complement to probing accuracy, particularly when comparing architectures with different internal signal scales.

5.5 Limitations

Single MoE model family. We study only OLMoE. Generalization to Mixtral (top-2 of 8), DeepSeek-MoE (fine-grained experts), or Qwen-MoE is open. The output dilution mechanism predicts that higher-sparsity architectures (lower k/N ratio) will show greater fragility, but this has not been tested.

Mean-pooling approximation. Per-expert probing and the perturbation experiments operate on mean-pooled representations, collapsing the sequence dimension. This approximation is standard in probing studies but may mask per-token routing effects. The router’s actual operation is per-token, not per-sequence.

Linear probes only. As in companion work, all probes are linear. Nonlinear probes (MLP classifiers) might extract moral features from the small-scale MoE output more effectively, potentially reducing the apparent fragility gap. However, the output scale measurement (§4.4) is independent of probe architecture.

Controlled but not identical comparison. OLMoE and OLMo-2 are from the same lab but differ in training data mix, hyperparameters, and training duration, not just architecture. Same-lab provenance minimizes but does not eliminate these confounds.

English and MFT only. The probing dataset covers English sentences grounded in Haidt’s Moral Foundations Theory. Moral encoding in other languages, moral frameworks, or culturally specific ethical norms is not tested.

6 Conclusion

We asked whether Mixture-of-Experts architectures create expert-level moral specialization, discrete modules that concentrate moral features and offer natural intervention points for alignment. The answer is no. In OLMoE-1B-7B, all 64 experts at every layer encode moral content with comparable accuracy (Gini < 0.03), and the router shows no preference for routing moral content to specific experts (maximum preference 1.8%).

This null on specialization led to a positive finding about architecture. MoE models are $4.2\times$ more fragile than dense models on moral probing despite matching on accuracy, and the mechanism is *output dilution*: the MoE block’s contribution to the residual stream is $74\times$ smaller in scale than the dense MLP’s, because sparse aggregation (top-8 of 64 experts) attenuates each expert’s contribution. The moral signal is present but encoded at a scale that is trivially overwhelmed by noise.

The checkpoint trajectory analysis (§4.5) deepens the null: the absence of specialization is not a late-training convergence but is present from the earliest available checkpoint (step 5K, 20B tokens). Moral encoding appears before the routing mechanism has fully matured, and the Gini coefficient of per-expert accuracy remains below 0.03 throughout training. Training does not create or destroy expert moral specialization; it was never there.

This finding refines the methodological program of companion work (Reblitz-Richardson, 2026). That work established fragility testing as a complement to probing accuracy for tracking alignment depth during pre-training. The present work shows that the gap between probing accuracy and fragility is not just a temporal phenomenon (fragility resolving after accuracy saturates) but an architectural one: MoE’s sparse aggregation creates a permanent structural fragility that no amount of training can resolve without changing the aggregation mechanism.

For future work, the output dilution mechanism makes specific predictions. Models with higher sparsity (lower k/N ratio) should show greater fragility. Fine-tuning should produce larger fragility shifts in MoE than in dense models of comparable active size. And MoE architectures that aggregate expert outputs differently (concatenation, attention-based mixing, or denser routing) should show correspondingly different fragility profiles.

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Appendices

Supplementary material. Sections referenced from the main paper as “Appendix A”–“Appendix D”.

A Probing dataset construction

The 240-pair moral probing dataset used throughout this paper is a deterministic subsample of a 1,200-pair dataset constructed per published quality guidelines (`DATASET_GUIDELINES.md`) with LLM-assisted filtering. The pipeline and full dataset are described in companion work (Reblitz-Richardson, 2026); we summarize the construction here for self-containment.

A.1 Seed extraction and pair generation

The 1,200-pair parent dataset covers six Moral Foundations Theory foundations (Graham et al., 2013, Haidt, 2012): care/harm, fairness/cheating, loyalty/betrayal, authority/subversion, sanctity/degradation, and liberty/oppression (200 pairs per foundation). Each pair consists of a moral sentence and a matched neutral sentence that preserves syntactic structure and topic domain while removing moral content. For example:

Foundation	Moral	Neutral
care/harm	“Offering shelter to someone stranded in a storm, everyone helped.”	“Offering directions to someone lost in a new city, everyone helped.”
fairness	“The manager promoted the most qualified candidate despite personal ties.”	“The manager promoted the candidate who had applied first.”
loyalty	“She reported her company’s illegal dumping to protect the community.”	“She reported her company’s quarterly earnings to the board.”

Neutral sentences are generated with LLM assistance and filtered for naturalness and moral neutrality of the neutral side.

A.2 Automated validation gates

Pairs pass length-ratio gates (≤ 1.5 ratio), keyword filtering (no explicit moral keywords in neutral sentences), and deduplication. The 1,200-pair dataset is released alongside the companion paper.

A.3 Subsampling

240 pairs (40 per foundation) are subsampled from the 1,200-pair parent with a deterministic seed (42). The subsample is split 80/20 into 192 training pairs (384 texts) and 48 test pairs (96 texts), with foundation balance preserved across splits.

A.4 Dataset identity across experiments

All experiments in this paper use the identical 240-pair subsample. The dense-vs-MoE comparison (§4.1), per-expert probing (§4.2), routing analysis (§4.3), and output scale comparison (§4.4) all process the same input texts, ensuring any observed differences are architectural rather than data-driven.

B Per-expert probing details

B.1 Full accuracy statistics by layer

The following table reports the full per-expert probe accuracy distribution across all 16 layers. “Above 90%” counts experts whose binary moral probe exceeds 90% accuracy on the 96-text test set; “Below 60%” counts experts near chance.

Layer	Mean	Std	Min	Max	Gini	>90%	<60%
0	0.825	0.031	0.760	0.896	0.021	0	0
1	0.825	0.032	0.719	0.896	0.021	0	0
2	0.857	0.034	0.750	0.948	0.022	7	0
3	0.844	0.035	0.740	0.917	0.023	2	0
4	0.880	0.035	0.792	0.938	0.022	20	0
5	0.905	0.030	0.854	0.969	0.019	39	0
6	0.910	0.032	0.833	0.979	0.020	39	0
7	0.902	0.029	0.844	0.969	0.018	31	0
8	0.908	0.026	0.865	0.979	0.016	36	0
9	0.914	0.026	0.854	0.958	0.016	47	0
10	0.920	0.028	0.844	0.958	0.017	49	0
11	0.927	0.029	0.854	0.990	0.018	54	0
12	0.930	0.027	0.833	0.979	0.016	57	0
13	0.927	0.030	0.854	0.979	0.018	53	0
14	0.930	0.028	0.844	0.990	0.017	55	0
15	0.916	0.033	0.823	0.969	0.020	49	0

1,020 of 1,024 probes (64 experts $\times$ 16 layers) exceed 75% accuracy (four early-layer probes at layers 1–3 reach 72–75%). No expert at any layer falls below 60%. The uniformity is striking: the Gini coefficient never exceeds 0.023 at any layer.

B.2 Gini coefficient interpretation

The Gini coefficient measures inequality in a distribution, ranging from 0 (perfect equality) to 1 (maximum inequality). For 64 experts, a Gini of 0.023 means the ratio of the best expert’s accuracy to the worst expert’s accuracy is approximately 1.3:1. For comparison:

- **No specialization** (observed): Gini 0.016–0.023
- **Mild specialization** (hypothetical): Gini 0.05–0.15, with a cluster of 5–10 “moral experts” clearly separated from the rest
- **Strong specialization** (hypothetical): Gini > 0.20 , with moral features concentrated in 2–5 experts and others near chance

The observed Gini values are an order of magnitude below even “mild specialization,” confirming that MoE architecture does not induce moral feature concentration.

B.3 Router analysis details

The router moral preference is computed as the difference in mean router logit between moral and neutral inputs, averaged across tokens. The maximum preference across all 64 experts and 16 layers is 1.8%, indicating near-complete content agnosticism. The router’s top-8 expert selection frequencies for moral and neutral inputs differ by less than 0.5% at every layer.

C Output scale measurement methodology

C.1 Hooking strategy

To measure feedforward output scale, we register forward hooks on the MLP module at each layer for both OLMoE and OLMo-2. The hook captures the module’s output *before* residual addition; this is the feedforward block’s contribution to the residual stream, isolated from the accumulated residual.

For OLMoE, `model.model.layers[1].mlp` returns a tuple (`aggregated_output`, `router_logits`); we capture the first element. For OLMo-2, `model.model.layers[1].mlp` returns the MLP output tensor directly.

C.2 Scale metric

We report the standard deviation of the feedforward output across all test texts (100 texts, drawn from the first 50 training pairs):

$$\text{output_std}_l = \text{std} \left(\left\{ \text{mean_pool}(\text{FFN}_l(x_i)) \right\}_{i=1}^{100} \right)$$

where `mean_pool` averages across the sequence dimension. This measures the *variability* of the feedforward output across inputs, i.e., the scale of the signal that the feedforward block contributes to the residual stream.

C.3 Per-layer output scale comparison

Layer	OLMoE FFN std	OLMo-2 MLP std	Ratio (OLMo/OLMoE)
0	0.003	0.45	167×
1	0.014	0.36	25×
2	0.051	0.27	5×
3	0.008	0.28	34×
4	0.008	0.33	41×
5	0.003	0.29	108×
6	0.003	0.29	87×
7	0.003	0.35	100×
8	0.008	0.46	60×
9	0.013	0.59	44×
10	0.011	0.57	51×
11	0.012	0.84	72×
12	0.018	1.07	61×
13	0.020	2.22	111×
14	0.040	4.68	117×
15	0.096	8.78	91×

The ratio varies considerably across layers (5× at layer 2 to 167× at layer 0), with a mean of 74× as reported in the main text. The lowest ratio at layer 2 reflects an unusually large MoE output at that layer, possibly due to early-layer representational adjustments.

C.4 Why the ratio varies across layers

The per-layer variation does not follow a simple monotonic pattern. The OLMo-2 MLP output grows from 0.28 (layer 2) to 7.99 (layer 15), spanning approximately 29×. The OLMoE aggregated output also grows but with more variability, spanning from 0.003 (layer 0) to 0.089 (layer 15). The ratio thus reflects both the growth rate difference and the layer-specific routing and aggregation dynamics of the MoE block.

C.5 Relationship to fragility

The output scale gap explains the fragility gap mechanistically. Gaussian noise $\mathcal{N}(0, \sigma^2 I)$ added to the full hidden state after the feedforward block perturbs both the residual and the feedforward contribution. Because the residual dominates the hidden state norm, the noise is calibrated to the residual scale. For the dense model, the MLP output is at a comparable scale to the residual, so the noise must be substantial to disrupt it. For the MoE model, the aggregated output is $74\times$ smaller, so noise that barely affects the residual already overwhelms the MoE contribution.

D Reproducibility

D.1 Hardware

All experiments run on a single MacBook Pro M4 Pro:

- 12-core CPU (8 performance + 4 efficiency)
- 24 GB unified memory (CPU and GPU share)
- M4 Pro GPU accessed via PyTorch MPS backend
- macOS (Darwin 25.x)

No GPU cluster, no CUDA. Total runtime across all experiments is approximately 2.5 hours of MPS compute time:

- Experiments 1+2 (per-expert probing + routing analysis): ~3 min
- Experiment 3 (component perturbation): ~15 min
- Output scale comparison: ~5 min
- Dense-vs-MoE layer probing (Experiment 5): ~10 min
- Experiment 4 (checkpoint trajectory, 11 checkpoints): ~1.5 hr

Model download time is not included; each OLMoE checkpoint is approximately 14 GB.

D.2 MPS compatibility patch

OLMoE’s router uses `torch.histc` for token counting, which is not implemented for integer tensors on MPS or CPU backends. We apply a minimal monkey-patch that casts to float and falls back to CPU for this single operation:

```
_orig_histc = torch.histc
def _histc_mps_fallback(input, bins=100, min=0, max=0):
    if input.device.type == "mps" or not input.is_floating_point():
        return _orig_histc(input.cpu().float(), bins, min, max).to(input.device)
    return _orig_histc(input, bins, min, max)
torch.histc = _histc_mps_fallback
```

This patch is applied in all OLMoE experiment scripts and does not affect numerical results (the operation counts tokens per expert for load-balancing diagnostics, not for gradient computation).

D.3 Random seeds

Experiment	Seed(s)	Where set
Probing dataset split	42	deepsteer/datasets/ pipeline.py
Per-expert probes (Exp 1)	torch default	exp1_2_expert_probing.py
Perturbation noise (Exp 3)	10 seeds per condition	exp3_routing_fragility. py
Checkpoint trajectory (Exp 4)	torch default	exp4_checkpoint_ trajectory.py

Perturbation experiments in Experiment 3 average over 10 random seeds per noise level to reduce variance from individual noise realizations.

D.4 Model checkpoints

Model	Repo	Revision	Used for
OLMoE-1B-7B	allenai/OLMoE-1B-7B-0924	main	Exp 1–3, Exp 5
OLMoE-1B-7B (trajectory)	allenai/OLMoE-1B-7B-0924	step5000-tokens20B through step1200000-tokens5033B	Exp 4
OLMo-2 1B	allenai/OLMo-2-0425-1B	main	Exp 5, output scale comparison

Both models are base (non-instruct) checkpoints loaded in float16 precision with `low_cpu_mem_usage=True`.

D.5 Command-line invocations

All commands run from the project root:

```
# Experiments 1+2: Per-expert probing and routing analysis
python papers/2_moe_output_dilution/scripts/exp1_2_expert_probing.py

# Experiment 3: Component perturbation fragility
python papers/2_moe_output_dilution/scripts/exp3_routing_fragility.py

# Output scale comparison (OLMoE vs OLMo-2)
python papers/2_moe_output_dilution/scripts/output_scale_comparison.py

# Experiment 4: Checkpoint trajectory analysis
python papers/2_moe_output_dilution/scripts/exp4_checkpoint_trajectory.py

# Experiment 5: Dense vs MoE layer-level comparison
python papers/2_moe_output_dilution/scripts/exp5_dense_vs_moe.py
```

D.6 Software versions

- Python 3.13
- PyTorch (with MPS backend)
- HuggingFace transformers and datasets
- numpy, matplotlib, seaborn

Exact versions are pinned in `pyproject.toml`.

D.7 Output JSON schema

Each experiment produces a structured JSON summary file with full metadata (model name, revision, hyperparameters, per-layer results). Files are located in `papers/2_moe_output_dilution/outputs/` under experiment-specific subdirectories. All experimental scripts and output files are released alongside the paper.